

Research Article

Who's Next? Turn Anticipation in Dutch Preschoolers With and Without Developmental Language Disorder

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ABSTRACT

Purpose: Children with developmental language disorder (DLD) sometimes experience difficulties with turn-taking. Smooth turn-taking requires efficient use of linguistic information in real-time interaction. Children as young as 2 years of age already use linguistic cues to predict upcoming conversational structure; for example, they anticipate more speaker switches following questions than non-questions (question advantage), especially questions with the second-person pronoun (“you”) than questions with the first-person pronoun (“I”; pronoun advantage). We investigated whether children with DLD were less efficient than their typically developing (TD) peers in using these linguistic cues to predict conversational structure in real time.

Method: We tracked the gaze of 23 three-year-olds with DLD and 23 TD peers while they watched two characters in conversation. We compared the size and timing of the question advantage and the pronoun advantage in both groups of children.

Results: We find no evidence that the sizes of the question advantage and pronoun advantage differed between children with DLD and their TD peers. However, the timing of their anticipatory looks to the addressee differed; TD children already looked more at the addressee for questions than nonquestions while the speaker was still talking, while children with DLD began looking more at the addressee after the speaker finished their turn. In their use of the pronoun cue, both groups of children already looked more at the addressee while the speaker was still talking.

Conclusions: Three-year-olds with and without DLD thus use linguistic cues to predict conversational structure in real time. The outcomes suggest, however, that children with DLD may be slower in their predictions. This slower processing may contribute to the turn-taking difficulties sometimes observed in children with DLD.

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Children learn language in the context of conversation with others. Successful conversations are characterized by smooth transitions between two (or more) conversational partners' speaking turns (de Vos et al., 2015; Sacks et al., 1974; Stivers et al., 2009). Achieving such smooth transitions requires the ability to predict if and when a response is needed. Conversations with either too fast (interruptions) or too slow responses are perceived as less fluent by native speakers as compared to conversations with conventional timing (van Os et al., 2020). To

become good conversationalists, it is thus important that children learn to precisely time their turns.

Language proficiency plays an important role in the precise timing of turns (e.g., Casillas et al., 2016; Levinson, 2016; Lindsay et al., 2019). Even when they are merely observing conversations, children make real-time predictions about possible turn transitions (e.g., one speaker responding to another one). Their ability to do so relies on their linguistic knowledge; anticipatory looks to turn transitions increase with age and the amount of available linguistic information. Thus, we might expect that children with developmental language disorder (DLD), who have relatively low language proficiency, experience more difficulties in processing real-time cues to upcoming turn structure and

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so make fewer anticipatory looks to upcoming speakers compared to their typically developing (TD) peers. Yet we know surprisingly little about the turn-taking skills of children with DLD.

In the present study, we aim to further examine the real-time processing of linguistic cues to upcoming turn structure in a group of children with DLD, who have difficulties learning language, and compare their anticipation of upcoming turn structure to a group of age-matched TD peers. A better understanding of the real-time anticipation of upcoming turn structure in children with DLD is important as it has both theoretical and clinical implications: It may help to explain why (pre)school children with language difficulties take fewer conversational turns with their peers or parents as compared to their TD peers (Blom et al., 2023; Hadley & Rice, 1991; Mohan et al., 2021).

From very early on, TD children engage in vocal turn-taking. Already at around 2 months of age, they make and respond to requests. These requests and responses (vocalizations) are timed remarkably well; infants' nonlinguistic vocalizations in response to a vocalization by their mother are often separated by less than 50 ms (Gratier et al., 2015). As they begin to produce linguistic forms, children's timing initially slows down (Gratier et al., 2015; Hilbrink et al., 2015), but at around 9 months of age, their responses get faster again (Hilbrink et al., 2015). Despite this, it takes children a long time (at least up until the age of 6 years, likely later) before they respond as quickly as adults in spoken conversation (Casillas et al., 2016; Nguyen et al., 2022; Stivers et al., 2018). Several authors have suggested that these fluctuations in children's timing of responses can be linked to different stages in children's social and linguistic development (Casillas et al., 2016; Hilbrink et al., 2015; Levinson, 2006).

Casillas et al. (2016), for example, observed a positive relation between the linguistic complexity of a child's answer and the time it takes a child to answer: The more complex the answer, the longer it takes a child to initiate a response. The authors also argue that older children do not necessarily respond faster than younger ones, given that the complexity of children's answers increases in parallel with their linguistic development and age. Inspired by Bögels et al. (2015), a study by Lindsay et al. (2019) showed that children's response times are impacted not only by the complexity of the response but also by the predictability of the questions. Here, predictability was manipulated by varying the moment at which informative material was provided: Questions contained either early informative material (e.g., "Which character, also called 007, appears in the famous movies," with the informative material underlined) or late informative material (e.g., "Which character from the famous movies is also called 007"; example taken from

Lindsay et al., 2019). As expected, preschoolers (3–6 years of age) responded faster to questions with early informative material as compared to questions with later informative material. It thus seems that linguistic proficiency and effective coordination between language comprehension and language production play an important role in children's accurate timing of conversational turns.

Further evidence that linguistic proficiency plays a role in children's accurate timing of conversational turns comes from experimental studies on children's real-time anticipation of conversational structure. In these studies, children watched videos of two characters in conversation and, while they watched the conversations, their anticipatory gaze switches from the speaker to the addressee (i.e., nonspeaking character) were measured. Consistently, these studies showed that the proportion of anticipatory gaze switches increased with age, potentially because children's developing linguistic knowledge allows them to make better use of the available linguistic cues to conversational structure (lexicosyntactic, prosodic, semantic; Casillas & Frank, 2017; Keitel et al., 2013; Lammertink et al., 2015). For example, Lammertink et al. (2015) showed that 2-year-old Dutch- and English-speaking children anticipated more speaker changes for utterances that were syntactically and prosodically complete versus utterances that were incomplete at one of these levels. Other studies have shown that from around 2 years of age, children start to anticipate more turn transitions for utterances that are linguistically marked for questions (e.g., *wh*-words, subject–verb inversion, rising intonation) as compared to utterances that are not linguistically marked for questions (Casillas & Frank, 2017; Lammertink et al., 2025).

Recent work by Lammertink et al. (2025) suggests that anticipatory looks to upcoming speakers can arise from fleeting but informative linguistic cues in the unfolding utterance. To test this idea, they showed children (aged 1;0, 2;0, 3;0, and 4;0 [years;months]) and adults a series of dialogues in Dutch with polar questions that either had "you" or "I" as the subject (e.g., "Are you sleepy?" vs. "Am I sleepy?"), finding greater anticipatory looks for "you" questions in adults and children who were aged 2;0 and older. Even though the subject pronoun was not phonetically salient in the stimuli, it was predicted to impact participants' predictions due to a reliable association between "you" and the act of asking questions. The findings showed evidence that participants have a heuristic for predicting upcoming talk based on individual linguistic cues (in that case, the subject pronoun). Critically, this anticipatory advantage for the second-person pronoun was only present when participants could understand other parts of the linguistic signal. That is, when the dialogues were shown to participants in a grammatically legal nonsense language (Jabberwocky Dutch), there was no

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“you” advantage, suggesting that the heuristic “you” cue must be integrated into real-time lexicosemantic comprehension before participants can realize any anticipatory advantage.

AQ7 *Turn-Taking in Children With DLD*

Despite the close and positive relationship between turn-taking and typical language development (Creaghe et al., 2021; Donnelly & Kidd, 2021; Katus et al., 2024; Sundqvist et al., 2022), we know little about turn-taking—particularly the relationship between language proficiency and turn-taking—in children with DLD. A recent systematic review on parental input and its relationship with language outcomes in children with (suspected) DLD by van Witteloostuijn et al. (2025) describes only two studies addressing this relationship. Blom et al. (2023) found that the number of conversational turns between children with DLD ($M_{\text{age}} = 42$ months) and their parents was positively associated with children’s receptive vocabulary scores ($r = .33$), but not with children’s expressive vocabulary or grammar scores. In a slightly older group of children with DLD ($M_{\text{age}} = 57$ months), and using a more qualitative measure of turn-taking, Toma and Orsolini (2013) observed a trend in which mothers of children with lower linguistic abilities (i.e., lower mean length of utterance scores) were more likely to be engaged in repair practices; particularly, they were more likely to take over the child’s responsibilities in making conversational repair. Naturalistic recordings of children with language difficulties (including DLD, speech impairments, and receptive and expressive language disorder) that also focus on quantitative aspects of turn-taking show that children with these language difficulties take fewer conversational turns in interactions with their peers and parents as compared to their TD peers (Blom et al., 2023; Hadley & Rice, 1991; Mohan et al., 2021). Studies exploring more qualitative aspects of interactional language suggest that, despite taking fewer turns in conversation, children with DLD are generally aware of the interactional norms of conversation and its sequential organization (Bishop & Adams, 1989; Justice & Kaderavek, 2003; Loveland et al., 1988; Moseley, 1990; Rice et al., 1991; Stojanovik, 2006; van Balkom et al., 2010; but see Craig & Evans, 1989; Mohan et al., 2021). The contributions that children with DLD make to the conversations, however, may be less complex and more often nonverbal (van Balkom et al., 2010). Together, these findings suggest that the reduced number of conversational turns in children with DLD cannot be explained by difficulties with general pragmatic aspects of turn-taking. In the present study, we therefore explore whether DLD instead impacts turn-taking behaviors via the real-time processing of linguistic cues needed for conversation. Specifically, for children with DLD, we wondered whether their lower language proficiency may hinder

conversational skills in two key ways: less available linguistic knowledge in speech production and comprehension, and difficulties in using fleeting cues key to the prediction of upcoming structure—cues essential for maintaining a smooth conversational flow.

In summary, both experimental and observational studies show that, as children grow older, their number of conversational turns increases (Donnelly & Kidd, 2021; Ferjan Ramírez et al., 2021; Katus et al., 2024) and their real-time anticipation of conversational structure becomes more efficient and adultlike (Casillas & Frank, 2017; Keitel & Daum, 2015; Lammertink et al., 2015, 2025). Furthermore, children’s developing pragmatic knowledge (i.e., their understanding that certain speech acts, such as greetings and questions, increase the likelihood for a needed response) and their developing linguistic knowledge (i.e., recognition of speech acts via linguistic cues) contribute to children’s developing turn-taking skills. However, it is unclear how and whether these processes contribute to turn-taking skills in children with DLD, compared to TD children. Previous studies have suggested that differences in pragmatic knowledge do not necessarily explain differences in turn-taking behavior between children with DLD and their TD peers (Bishop & Adams, 1989; Loveland et al., 1988; Moseley, 1990; Rice et al., 1991; Stojanovik, 2006; van Balkom et al., 2010; but see Craig & Evans, 1989). The present study therefore examines whether children with DLD and their TD peers differ in their processing of linguistic cues to questionhood and their likelihood of anticipating a response by the addressee while observing two characters in conversation.

The Present Study

Children with DLD experience difficulties with language across multiple domains, including the lexicon, morphosyntax, discourse, reading, and spelling (Leonard, 2014; McArthur et al., 2000). Despite heterogeneity in the problems observed across children with DLD, almost all children with DLD demonstrate problems with morphosyntax and phonological processing. We hypothesize that the latter makes it more difficult for children with DLD to use fleeting and sophisticated linguistic cues in their prediction of conversational structure. For example, children with DLD may struggle more than TD children in rapidly recognizing subject–verb inversion (morphosyntactic information) or a final rising contour as cues to questionhood. To test our hypothesis, the present study compared how children process different linguistic cues to questionhood: one set of cues associated with questions by “rule” (subject–verb inversion and rising intonation as standard markers of polar questions) and one cue that is only associated with questions by heuristic (a second-person subject pronoun;

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speakers are more likely to be asking about “you” [and telling about “I”]). The second kind of cue is merely heuristic in indicating questionhood because not all utterances with the second-person pronoun are questions, nor do all questions have a second-person pronoun (see also Lammertink et al., 2025).

We tested how these cues are processed in the real-time anticipation of conversational structure by 3-year-old Dutch children with and without suspected DLD. We use the term “suspected DLD” here, because DLD is commonly identified only at 4–5 years of age, and at this young age, it is difficult to distinguish between children with DLD and late talkers (e.g., Moyle et al., 2011). All participating children with suspected DLD were part of an early intervention program focused on language development, because they had been flagged as “at risk for DLD” by professionals, based on standardized language tests. In the Netherlands, most children with DLD start joining early intervention groups at around 3 years of age. This is why the present study focuses on 3-year-olds.

In the first set of analyses, we focused on children’s use of rule-associated cues to questionhood: subject–verb inversion and rising intonation. While children watched videos of two female characters in conversation, we (a) compared their likelihood of making anticipatory gaze switches from the speaking character to the addressee (nonspeaking character) for questions versus nonquestions and (b) determined the moment in time (in relation to the speaking character’s turn end and in relation to occurrence of the subject–verb inversion cue) at which children started to look at the addressee. In both cases, we expected to observe a larger question advantage (i.e., more anticipatory looks for questions than nonquestions and earlier gaze switches for questions than nonquestions) in TD children than in children with suspected DLD.

Second, to draw conclusions about children’s use of heuristic cues to questionhood—here, the subject pronoun used—we analyzed children’s behavior for questions with the second-person pronoun (you) versus questions with the first-person pronoun (I). We predicted a larger second-person pronoun advantage in terms of both (a) the likelihood of making an anticipatory look and (b) the timing of looks to the addressee for TD children as compared to children with suspected DLD.

In our third set of analyses, we conducted a more exploratory study on the link between language proficiency and the anticipatory behavior in children with suspected DLD. Specifically, we examined the correlations between children’s scores on the Schlichting tests for language comprehension and language production (Schlichting & Lutje Spelberg, 2010a, 2010b) and the question and pronoun advantages measured in the experimental data (i.e.,

from Analyses 1 and 2 above). We only examined these correlations in the children with suspected DLD, as we had no information on the language scores of the TD children.

Method

Participants

We recruited 27 toddlers with suspected DLD via special treatment groups for toddlers with DLD within Royal Dutch Kentalis ($n = 21$) and the Dutch Foundation of the Deaf and Hard-of-Hearing Child (NSDSK; $n = 5$). Before conducting our analyses, we excluded four children with suspected DLD for the following reasons: The speech therapist later reported that the child was diagnosed with attention-deficit/hyperactivity disorder ($n = 1$), we failed to obtain eye-tracking data ($n = 2$), or none of the child’s real-time fixations to the stimuli met our criteria to be considered an anticipatory look ($n = 1$; see the Data Pre-processing section). The remaining 23 children with suspected DLD were then roughly matched on age and gender¹ to preexisting data from 23 TD children who had participated in a prior study (Lammertink et al., 2025). These TD children were recruited from the participant pool of the Baby & Child Research Center Nijmegen.

For the final data set, the mean age of the children with DLD (17 boys, six girls) was 43 months ($SD = 4$ months, range: 34–50 months) and the mean age of the TD children (13 boys, 10 girls) was 43 months ($SD = 6$ months, range: 36–49 months). Three children with DLD but none of the TD children were being raised multilingually (i.e., their parents indicated that they spoke another language besides Dutch with their child at home; see the Discussion section and additional post hoc analyses in Supplemental Material S1 for an exploration of the implications of including these multilingual children). No child had parental-reported hearing loss, vision problems, neurological or syndromic disorders, or indication of cognitive disability. The children with suspected DLD were all diagnosed by a speech-language therapist as having language difficulties in the context of nonverbal intelligence within the normal range. For 22 (out of 23) children with DLD, we received scores for their word comprehension from the Schlichting Test for Language Comprehension (Schlichting & Lutje Spelberg, 2010a) and their word production from the Schlichting Test for Language Production (Schlichting & Lutje Spelberg, 2010b). Furthermore, for 21

¹In response to a reviewer request, we explored post hoc whether the unequal distribution of boys and girls across the two groups of children could have impacted the results. The outcomes of these additional analyses are reported in Supplemental Material S1 and provide no evidence that this is the case.

(out of 23) children with suspected DLD, we also received scores for their sentence production, from the Schlichting Test for Language Production. The mean percentile scores of the children with suspected DLD on the language tests were as follows: language comprehension (TBQ), 92 ($SD = 8$, range: 74–104); word production, 79 ($SD = 17$, range: 55–103); and sentence production, 75 ($SD = 7$, range: 64–89). Note that, at the group level, the average language comprehension percentile score for our group of children with DLD is relatively high (i.e., it falls within the average range of a standardized score between 85 and 115). We discuss this further in our Discussion section.

Materials

AQ16 The videos used in the present study were a subset of those used in Lammertink et al.'s (2025) study. There were eight videos, each featuring an 11-utterance dialogue about child-friendly topics (e.g., feeding and petting animals, visiting the pool, building a snowman). Among the 11 utterances in each dialogue, four utterances were polar (“yes/no”) questions (e.g., Do you like them?). All questions were marked with subject–verb inversion (e.g., Dutch: *Mag jij ze binnen zetten?* [Are you allowed to put them inside?]) and rising intonation. Across all dialogues, half of the questions had a first-person subject pronoun (*ik* [I]; e.g., *Zal ik ze dan buiten laten?* [Should I leave them outside?]), while the other half of the questions had a second-person subject pronoun (*jij* [you]; see example above).

AQ17 To address our first research question, we compared children's gaze behavior to questions versus nonquestions. In the original study by Lammertink et al. (2025), this comparison between questions and nonquestions was not part of the research question. In that study, the nonquestioning utterances ($n = 7$ of the total 11 utterances) served as filler items. For the present study, we selected, from each dialogue, four nonquestioning utterances (out of the available seven) as “nonquestion” target items to compare with the four “question” target items (see below for nonquestion selection criteria). The remaining three filler (nonquestioning) were still considered as filler items in the present analysis. Thus, in total, the children watched eight 11-utterance dialogues, each consisting of four question targets, four nonquestion targets, and three filler utterances. Across the whole set of eight dialogues, children thus heard 32 question targets and 32 nonquestion targets with the additional constraints that (a) question and nonquestion target utterances were equally likely to be followed by an actual turn switch (i.e., a response by the other, nonspeaking character) or by a turn continuation (i.e., the character who previously talked resumes talking) and (b) question and nonquestion target utterances were

equally divided over the two speaking characters, with a total of 16 question and 16 nonquestion targets each.

To address our second research question, we compared children's gaze behaviors to questions with the first-person pronoun versus second-person pronoun (as in Lammertink et al., 2025). In total, children heard 16 questions with the first-person pronoun and 16 questions with the second-person pronoun. Again, dialogues were structured such that, across the whole set of eight dialogues, the following constraints held: (a) Questions with the first-person pronoun and second-person pronoun were equally likely to be followed by a real turn switch or by a turn continuation, and (b) the first-person pronoun questions and second-person pronoun questions were equally divided over the two speaking characters, with a total of eight first-person pronoun and eight second-person pronoun questions each. A dialogue example is presented in Figure 1.

The dialogues were voiced by two adult female native Dutch speakers across multiple rounds of recording (for more details about the recording sessions, see Lammertink et al., 2025). In the final dialogue audio signal, each speaker's voice was assigned to a single channel (left or right), and all interutterance silences (which we refer to as “gaps” in the remainder of the article) were 500 ms long (see also Casillas & Frank, 2017; Lammertink et al., 2015).

Figure 1. Script for one of the eight dialogues. Bolded lines show target sentences. Dutch is shown in blue with loose translations to English in black.

A: *Ik heb twee konijnen gekregen.*

I got two rabbits.

Zul jij ze leuk vinden?

Do you like them?

B: *Ik vind konijnen lief.*

I think rabbits are sweet.

Ik heb zelf ook een konijn.

I also have a rabbit.

A: *Ik zal mijn konijnen ophalen.*

I'll go get my rabbits.

Neem ik dan mee naar binnen?

Should I bring them inside?

B: *Mag jij ze binnen zetten?*

Are you allowed to put them inside?

Mijn konijn mag nooit naar binnen.

My rabbit is never allowed inside.

Het huis wordt dan vies.

Then the house would get dirty.

A: *Zal ik ze dan buiten laten?*

Should I leave them outside?

Het is toch warm buiten.

It's warm out after all.

The final audio file for each dialogue was then integrated into a simple animation of two female characters talking (see Casillas, 2021, for relevant scripts). The characters featured in the dialogue were illustrated as a face-to-face pair of cartoons depicting pale-skinned girls with contrasting hair and shirt colors: a red-headed character wearing a purple shirt and a blonde character wearing a blue shirt (see Figure 2). We counterbalanced character location and dialogue order across children as follows: Children either saw the blond character on the left or the right, and the dialogues played in one of two orders: from Dialogue 1 to 8 or from Dialogue 8 to 1. Within a single test session, a participant saw only one version of a set of videos.

Procedure

The children with suspected DLD were tested in quiet rooms at their treatment group locations. Their eye movements were measured with a portable Tobii Pro X3-120 EPU that was directly attached (via a magnetic strip) to a laptop monitor (1920 × 1080 pixels). The videos were displayed on the monitor of the laptop, and the stereo audio was played over the speakers of the laptop. The experiment was conducted in Tobii Studio (3.4.8.1348).

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As described in Lammertink et al. (2025), the TD children were tested in quiet testing rooms within the Baby & Child Research Center Nijmegen. The eye movements of the TD children were measured with an EyeLink 1000, attached by an ergonomic arm to a table with the tracker computer. The videos were displayed on a monitor (1920 × 1080 pixels), and the audio was played over stereo speakers (Genelec 8010A Studio Monitor) placed just behind the screen to its left and right. The tracker was run from one computer using EyeLink's custom operating system (EyeLink 5.13), with the experiment itself run from a second computer using Presentation (Version 21.1, Build 09.05.19; Neurobehavioral Systems) and a video of the session recorded using OBS Studio (21.1.0) via a webcam

that was attached to the stimulus monitor and connected to a third computer.

All participants' parents signed a consent form prior to participation. For children with suspected DLD, this research was approved by the Radboud University Faculty of Arts Ethics Committee (21 U.015164). For the TD children, our research was approved by the Radboud University Social Sciences Ethics Committee (ECSW/MB 17 U.000160). For both groups of children, the task took children approximately 5 min to complete.

Data Preprocessing and Analysis

All data, scripts, and full model outcomes are publicly accessible in our Radboud Data Sharing Collection (Lammertink, Rowland, & Casillas, 2025). Before analyzing children's gaze (for both their proportion of anticipatory looks and their proportion of looks to the addressee), the raw data set was preprocessed to (a) replace unreliable tracker output with NA values (33.7% of datapoints for children with suspected DLD and 32.6% of the datapoints for TD children) and (b) remove individual (but complete) dialogues in which children attended to the screen for less than 25% of the total dialogue duration. The latter led to a total exclusion of 17 dialogues (corresponding to a total of 136 target utterances for Analysis 1 and 68 target utterances for Analysis 2) in both children with DLD and TD children.

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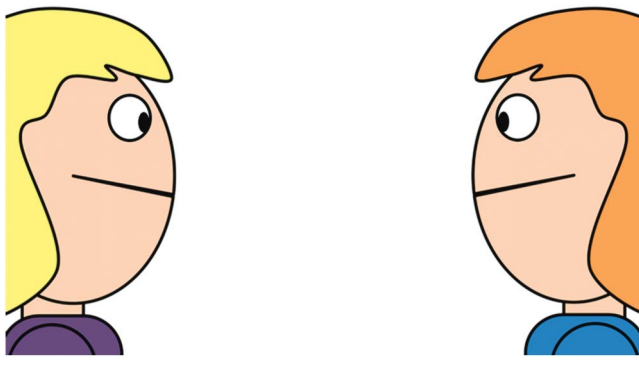
Likelihood of Anticipatory Gaze Switches

Following previous work (Casillas & Frank, 2017; Lammertink et al., 2015, 2025), our first analysis compared the likelihood of making an anticipatory gaze switch to the addressee (i.e., nonspeaking character) upon hearing questions versus nonquestions (RQ 1A) and upon hearing questions with the second-person pronoun versus questions with the first-person pronoun (RQ 2A) between children with and without suspected DLD. We set specific criteria to determine whether, at the end of each target turn, the participant switched their gaze from the speaker to the addressee in anticipation of a response or not. Our criteria for an anticipatory gaze switch were as follows: A participant makes an anticipatory gaze switch if they (a) were fixating on the current speaker within the last 300 ms of the speaker's turn for at least 100 ms and then (b) began fixating for at least 100 ms on the other (nonspeaking) character within the first 300 ms of the following utterance. This led to a total anticipatory window of 1,100 ms centered on the 500-ms silent gap following each of the target utterances (300-ms pregap + 500-ms gap + 300-ms postgap). If a child's gaze met these criteria, the look was counted as anticipatory and given a score of 1. If not, the gaze was not counted as anticipatory and given a score of 0. If children's gaze was not captured for a minimum of 100 ms

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Figure 2. Example of the visual display with the two characters.



during each subwindow of analysis (for Criteria a and b above), they were given a score of NA. There was one child with DLD who only received NA values following this latter criterion, and therefore, we excluded this child from analysis.

To compare the size of the question advantage and the size of the pronoun advantage, we ran two generalized mixed-effects regression models (package *lme4*, Version 1.1-33; Bates et al., 2015; R Core Team 2023, Version 4.3.1). The models fitted anticipatory gaze switch (1 = anticipatory switch, 0 = no anticipatory switch) as a function of the binary predictors: group (DLD vs. TD), target (RQ 1A: questions vs. non-questions; RQ 2A: second-person pronoun vs. first-person pronoun), and NextTurn (turn continuation by current speaker vs. turn transition from one speaker to another). In both models, the random-effects structure contained by-subject ($n = 46$) and by-item (RQ 1A: $n = 64$; RQ 2A: $n = 32$) random intercepts. Our effects of interest were contrast coded such that (a) a positive estimate for group indicated more anticipatory looks for TD children (+1/2) as compared to children with suspected DLD (−1/2) and (b) a positive estimate for target indicated more anticipatory looks for questions/second-person pronoun questions (+1/2) than nonquestions/first-person pronoun questions (−1/2).

Proportion of Looks to the Addressee Over Time

Our anticipatory gaze switch analysis (see the Likelihood of Anticipatory Gaze Switches section) tests whether looking behavior during the test trials is anticipatory (1) or nonanticipatory (0) but does not provide information on children's looking behavior over the course of a possible turn transition. The timing of anticipatory looks is critical to getting insight into real-time processing differences between groups, across conditions. Therefore, to investigate children's looking behavior over time, we carried out several cluster-based permutation analyses using the *eyetrackingR* package (Dink & Ferguson, 2015). In this second set of analyses, the dependent variable was children's proportion of looks to the addressee (i.e., other, nonspeaking character). Note that the cluster-based permutation analyses included all looks to the addressee, whether the look met our criteria for being anticipatory or not (see the Likelihood of Anticipatory Gaze Switches section). We analyzed children's looks to the addressee in two different time windows: The first window was the 1,100-ms window centered around the gap (the postutterance window; see the Likelihood of Anticipatory Gaze Switches section), and the second window was the 800-ms² window following the subject–verb inversion cue (i.e., start of the utterance, RQ 1) or the 700 ms³ following pronoun

onset (RQ 2; the postcue window). The first window was chosen because it aligns with the time window of the anticipatory gaze switch analysis (see the Likelihood of Anticipatory Gaze Switches section). The second window was chosen because Lammertink et al. (2025) showed that the second-person pronoun advantage in 4-year-olds already starts before the onset of the anticipatory gaze window, which makes sense given that the pronoun cue appears very early in the utterance. Here, we were interested to see whether this early looking pattern also holds for the question advantage, where the subject–verb inversion cue is presented immediately at the start of the utterance.

The data for the permutation analyses were preprocessed so that only those target utterances for which we had at least 25% valid looks from a child within our time window of interest were included. Additionally, individual datapoints (looks) were removed if they were recorded with a timestamp smaller than 0.0015 ms, which is far below the set sampling frequency of the trackers during data collection yet still occurred around 5% of the time in the TD data set (it never occurs in the DLD data set, which used a different tracker system). For the question advantage analyses and Time Window 1 (1,100 ms centered around the gap), these exclusion criteria led to an exclusion of 214 target utterances (in addition to the 136 target utterances that were already removed by the full dialogue exclusion; see the Data Preprocessing and Analysis section) in children with DLD. In TD children, this preprocessing led to an exclusion of an extra 211 individual target trials (in addition to the 136 target utterances that were already removed by the full dialogue exclusion) and an exclusion of 5.8% of individual datapoints because of small timestamps. For Time Window 2 (800 ms following subject–verb inversion), this preprocessing led to an exclusion of an additional 187 target trials in children with DLD. In TD children, this led to an exclusion of an additional 190 target trials and an exclusion of 5.9% of individual datapoints because of small timestamps. For the pronoun advantage analyses and Time Window 1 (1,100 ms centered around the gap), preprocessing led to an exclusion of 106 target trials (in addition to the 68 target utterances that were already excluded by the full dialogue exclusion) in children with DLD. In TD children, this led to an exclusion of 104 target trials (in addition to the 68 target utterances that were already removed by the full dialogue exclusion) and an exclusion of 5.8% of additional individual datapoints because of small timestamps. For Time Window 2 (700 ms following pronoun onset), preprocessing led to an exclusion of 101 additional target trials in children with DLD. In TD children, this led to an exclusion of 108 additional target trials and an exclusion of 5.8% of additional individual datapoints because of small timestamps.

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²This corresponds to the shortest total trial duration of the 64 target utterances.

³This corresponds to the shortest duration from trial onset to gap across the 32 target utterances.

After exclusions, the remaining tracker datapoints were split into 50-ms time bins. For each time bin, we calculated a t statistic to determine whether children looked significantly more to the addressee for questions than non-questions (question advantage; RQ 1B) or for questions with the second-person pronoun versus first-person pronoun (pronoun advantage; RQ 2B). Then, adjacent time bins showing these significant differences by target type were clustered together, and the t values of the bins were summed. To correct for false positives, the analyses included a second step in which we reshuffled the data 1,000 times to estimate how likely the observed clusters and summed statistics were if the data were randomly generated. Note that, because we use the t statistic from these cluster-based permutation analyses, we can only compare one predictor at a time. Consequently, our cluster-based permutation analyses only compared the timing of the question advantage (RQ 1B) and pronoun advantage (RQ 2B) within—and not between—the two groups of children.

Exploratory Analyses: Correlations Between Language Proficiency and the Question and Pronoun Advantages

Finally, given that we had language scores on the Schlichting tests for language production and language comprehension for most of the children with suspected DLD, we conducted a third exploratory analysis of the correlations between the size of the children's question and pronoun advantages as well as their percentile scores for language production (both word production and sentence production) and language comprehension. For each individual child with suspected DLD, we computed their question and pronoun advantages by subtracting their proportion of anticipatory looks to nonquestion/questions with the first-person pronoun from their proportion of anticipatory looks to questions/questions with the second-person pronoun. Correlations between language proficiency and the size of question and pronoun advantages could not be computed for the TD children, because we have no language scores for these children.

Results

Question Advantage

Likelihood of Anticipatory Gaze Switches

F3 Figure 3 shows the proportion of anticipatory gaze switches for questions (purple) and nonquestions (light blue) in children with suspected DLD (left) and TD children (right). The model estimated that, averaged over both groups of children, the likelihood that children anticipate a turn transition is 5.3 times larger for questions than for nonquestions (log-odds estimate of target = 1.7,

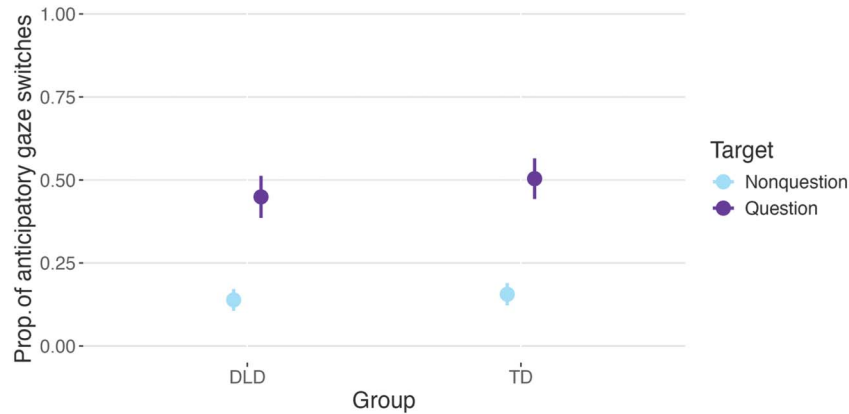
odds ratio [OR] = 5.3; z = 7.8, p < .001; 95% Wald confidence interval [CI] OR [3.5, 8.0]; see Table 1). Overall, children thus show the expected question advantage (Casillas & Frank, 2017; Lammertink et al., 2015). The model also estimated that the question advantage is numerically larger for TD children as compared to children with DLD (log-odds estimate of target:group = 0.092, OR = 1.1), but this estimation is statistically not significantly different from zero (z = 0.36, p = .72; 95% Wald CI OR [0.70, 1.8]); the findings do not support the idea that the size of the question advantage differs between children with and without suspected DLD.

Proportion of Looks to the Addressee Over Time

Figure 4 shows the mean proportion of looks to the addressee (nonspeaking character) for questions and non-questions during our first time window, which starts 300 ms before the gap and runs until 300 ms after the gap (i.e., the similar 1,100-ms time window as used in the proportion of anticipatory looks analysis). It shows the proportion of looks to the addressee in children with suspected DLD (see Figure 4A) and TD children (see Figure 4B). The permutation analyses detected one cluster where children with DLD looked more at the addressee for questions than nonquestions, and this cluster started 150 ms after gap onset and ran until the end of our time window (i.e., 300 ms after gap offset; duration = 650 ms; sum t = -80.3, p < .001). In TD children, the analyses detected that children started to look more at the addressee for questions than nonquestions 150 ms before gap onset until the end of our time window (i.e., 300 ms after gap offset; duration = 950 ms, sum t = -85.0, p < .001). More specifically, TD children already start to look at the addressee while the speaking character is still talking (i.e., before gap onset), whereas children with suspected DLD start to look at the addressee right after the current speaking character stopped talking.

Figure 5 shows the mean proportion of looks to the addressee for questions and nonquestions during our second time window, which lasts for 800 ms following the subject-verb inversion cue (i.e., trial onset). Again, it shows the proportion of looks to the addressee in children with suspected DLD (see Figure 5A) and TD children (see Figure 5B). In children with DLD, the permutation analyses detected no clusters where children looked more at the addressee for questions than nonquestions. In TD children, the analyses detected four clusters in which children looked more at the addressee for questions than nonquestions (see Table 2), but looks to the addressee were only significantly different from zero in one cluster, starting 500 ms after utterance onset and running until 700 ms after utterance onset (duration: 200 ms; sum t = -11.0, p = .040). Thus, TD children (at least temporarily) start to look more at the

Figure 3. Children's mean proportion (Prop.) of anticipatory gaze switches for questions (purple) and nonquestions (light blue), split by group. Children with developmental language disorder (DLD) on the left and typically developing (TD) children on the right.



addressee for questions than nonquestions very early on in the utterance, about 500 ms after trial onset, suggesting that they use subject–verb inversion as an early cue to question-hood in unfolding conversations.

Pronoun Advantage

Likelihood of Anticipatory Gaze Switches

F6 Figure 6 shows the proportion of anticipatory gaze switches for questions with a second-person pronoun (orange) and a first-person pronoun (green) in children with suspected DLD (left) and TD children (right). The model estimated that, averaged over both groups of children, the likelihood that children anticipate a turn

transition is 2.7 times larger for questions with the second-person pronoun as compared to questions with the first-person pronoun (log-odds estimate of target = 1.0, $OR = 2.7$; $z = 4.6$, $p < .001$; 95% Wald CI $OR [1.8, 4.1]$; see Table 3). Overall, children thus show a pronoun advantage, as expected (Lammertink et al., 2025). The model also estimated that the pronoun advantage is larger for TD children as compared to children with suspected DLD (log-odds estimate of target:group = 0.17, $OR = 1.2$), but this estimate is statistically not significantly different from zero ($z = 0.50$, $p = .62$; 95% Wald CI $OR [0.60, 2.3]$); the findings do not support the idea that the size of the pronoun advantage differs between children with and without suspected DLD.

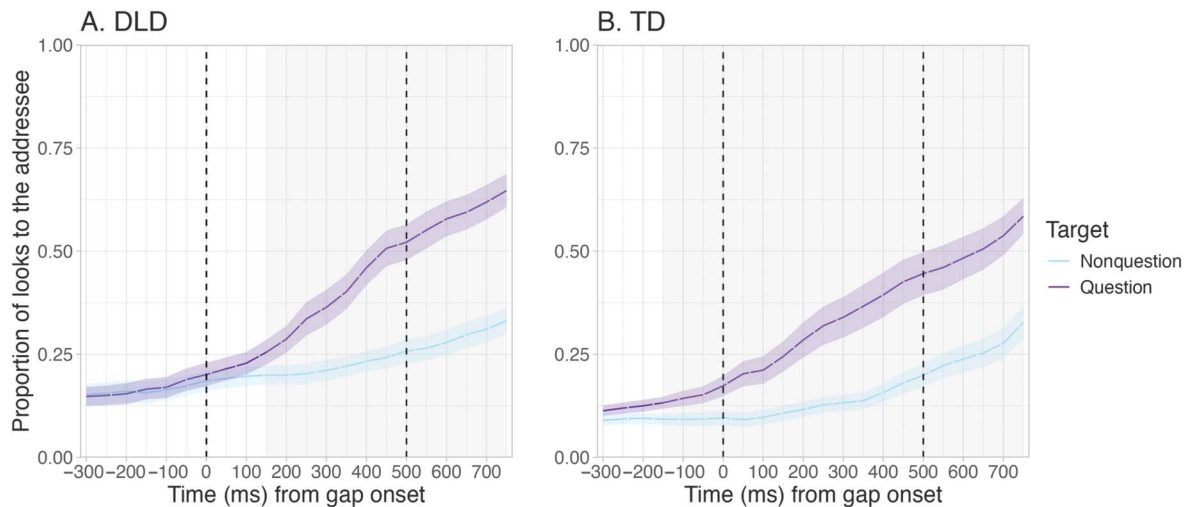
T3
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Table 1. Model outcomes and Wald confidence intervals (CIs) for the size of the question advantage for the likelihood of anticipatory gaze switches (generalized linear mixed-effects model; 1,652 observations).

Random effects					SD
Subject ($n = 46$)					
Intercept					0.88
Item ($n = 64$)					
Intercept					0.67
Fixed effect	Estimate (log-odds)	Standard error (log-odds)	95% CI (odds ratio)	z	p
Intercept	−0.92	0.17	[0.3, 0.6]	−5.1	< .001
Target (question vs. nonquestion)	1.7	0.21	[3.5, 8.0]	7.7	< .001
Group (DLD vs. TD)	0.18	0.30	[0.7, 2.1]	0.87	.55
NextTurn (transition vs. continuation)	0.35	0.21	[0.9, 2.2]	1.8	.092
Target × Group	0.092	0.25	[0.7, 1.8]	0.26	.73
Target × NextTurn	0.18	0.42	[0.5, 2.7]	0.42	.67
Group × NextTurn	−0.059	0.25	[0.6, 1.5]	0.044	.81
Target × Group × NextTurn	0.55	0.50	[0.6, 4.6]	1.06	.28

Note. DLD = developmental language disorder; TD = typically developing.

Figure 4. Children's proportion of looks to the addressee for nonquestions (purple) and questions (light blue) from 300 ms before the gap to 300 ms after the gap. The area between the two solid lines (from 0 to 500 ms) indicates the 500-ms gap (interutterance silence). The light gray shading indicates time bins that were found to be statistically significant in the permutation analysis. (A) Gaze behavior of the children with developmental language disorder (DLD). (B) Gaze behavior of the typically developing (TD) children.



Proportion of Looks to the Addressee Over Time

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Figure 7 shows the mean proportion of looks to the addressee for questions with the second-person pronoun and questions with the first-person pronoun during our first time window, which starts 300 ms before the gap and runs until 300 ms after the gap. In children with DLD, the permutation analyses detected one cluster during which the children looked more at the addressee for questions with the second-person pronoun as compared to

questions with the first-person pronoun, and this cluster covered the full time window (duration = 1,100 ms; sum $t = -64.4$, $p < .001$). In TD children, the detected cluster also covered almost the full time window—from 350 ms before the gap till 300 ms after the gap—during which TD children looked more at the addressee for questions with the second-person pronoun as compared to questions with the first-person pronoun (duration = 1,050 ms; sum $t = -102.0$, $p < .001$). Visual inspection of Figure 7 suggests

Figure 5. Children's proportion of looks to the addressee for nonquestions (purple) and questions (light blue) in the 800 ms following the subject-verb inversion cue (i.e., from utterance onset). The light gray shading indicates time bins that were found to be statistically significant in the permutation analysis. (A) Gaze behavior of the children with developmental language disorder (DLD). (B) Gaze behavior of the typically developing (TD) children.

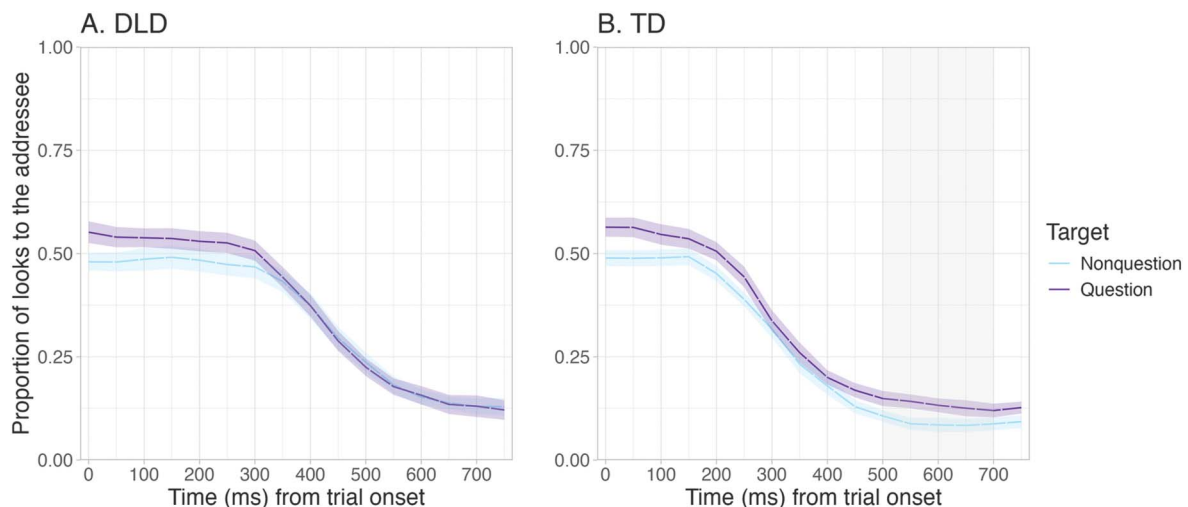


Table 2. Outcomes of the cluster-based permutation analysis for the question advantage following 800 ms after subject–verb inversion cue (i.e., utterance onset).

TD children			
Time cluster	Sum statistic	Time range in ms	Probability
1	−4.41	0.0–100	.14
2	−2.09	250–300	.26
3	−11.0	500–700	.040
4	−2.14	750–800	.24

Note. TD = typically developing; ms = milliseconds.

that both groups of children already start to look more at the addressee for questions with the second-person pronoun before the start of our analysis window (i.e., more than 300 ms before the gap), suggesting this looking behavior is triggered earlier (e.g., immediately following presentation of the pronoun itself). We explore this early, cue-driven looking in our second time window analysis below (see Figure 8).

Figure 8 shows the mean proportion of looks to the addressee for questions with the second-person pronoun and questions with the first-person pronoun during our second time window, which follows 700 ms after pronoun onset. In the children with DLD, the permutation analyses detected one cluster where children with DLD looked more at the addressee for questions with the second-person pronoun than clusters with the first-person pronoun. This cluster started 550 ms after pronoun onset and ran until 700 ms after pronoun onset but was not statistically significantly different from zero (duration = 150 ms; sum $t = -6.64$, $p = .097$). In TD children, the permutation analysis also detected one cluster in which children looked more at the addressee for questions with the second-person pronoun as compared to questions with the first-person

pronoun. This cluster was statistically significantly different from zero and started 500 ms after pronoun onset and ran until 700 ms after pronoun onset (duration = 200 ms; sum $t = -11.54$, $p = .032$).

Taken together, the outcomes of the permutation analyses suggest that both groups of children start to look at the addressee for questions with the second-person pronoun more than for questions with the first-person pronoun before gap onset, and for TD children, this is approximately 500 ms after pronoun onset. It remains unclear when, after pronoun onset, children with DLD start to look more at the addressee for questions with the second-person pronoun, given the absence of evidence for a statistically significant cluster.

Exploratory Analyses

Given that there was considerable variation in the size of the individual question advantages of children with DLD ($M = 0.28$, $SD = 0.19$; range: -0.13 to 0.67), we further explored correlations between the size of children's question advantages and their language scores. We observed a trend in which children with larger question advantages scored better on the Schlichting Test for Language Comprehension (Pearson $r[20] = .40$), but the correlation was statistically not significantly different from zero ($p = .06$, 95% CI $[-0.20, +0.71]$). Children with larger question advantages also scored higher on the Schlichting Test for Word Production (Pearson $r[20] = .073$), but again, the correlation was not statistically significantly different from zero ($p = .75$, 95% CI $[-0.46, +0.48]$). Finally, children with larger question advantages scored lower on the Schlichting Test for Sentence Production (Pearson $r[19] = -.16$) but this correlation was also not statistically significantly different from zero ($p = .49$, 95% CI $[-0.55, +0.29]$).

Figure 6. Children's mean proportion (Prop.) of anticipatory gaze switches for questions with the first-person pronoun (green) and questions with the second-person pronoun (orange) split by group. Children with developmental language disorder (DLD) on the left and typically developing (TD) children on the right.

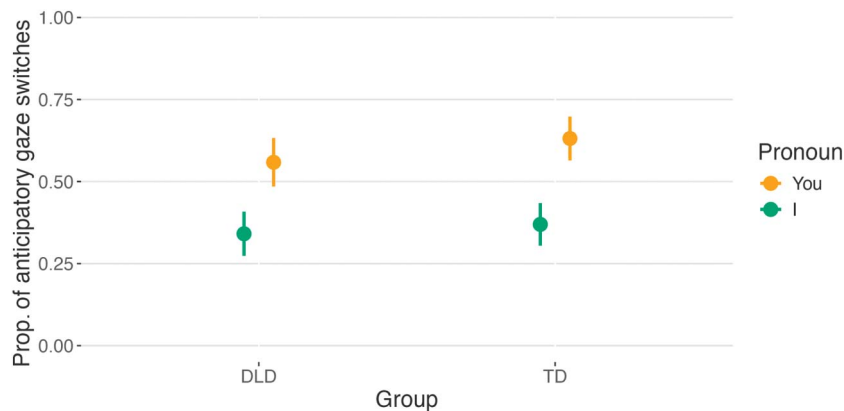


Table 3. Model outcomes and Wald confidence intervals for the size of the pronoun advantage for the likelihood of anticipatory looks (generalized linear mixed-effects model; 814 observations).

Random effects					SD
Subject ($n = 46$)					
Intercept					1.0
Item ($n = 32$)					
Intercept					0.38
Fixed effect	Estimate (log-odds)	Standard error (log-odds)	95% CI (odds ratio)	z	p
Intercept	−0.094	0.19	[0.7, 1.4]	−0.50	.61
Target (you vs. I)	1.00	0.22	[1.8, 4.2]	4.6	< .001
Group (DLD vs. TD)	0.22	0.35	[0.7, 2.7]	0.63	.53
NextTurn (transition vs. continuation)	0.45	0.21	[1.1, 2.5]	2.1	.037
Target × Group	0.17	0.33	[0.6, 2.4]	0.50	.62
Target × NextTurn	0.52	0.43	[0.7, 3.9]	1.2	.22
Target × NextTurn	0.27	0.33	[0.7, 2.7]	0.83	.40
Target × Group × NextTurn	−0.39	0.65	[0.2, 2.5]	−0.60	.55

Note. CI = confidence interval; DLD = developmental language disorder; TD = typically developing.

Finally, we also explored whether differences in the size of the pronoun advantage in children with DLD ($M = 0.11$, $SD = 0.35$; range: -0.71 to 0.67) correlated with their language scores. However, none of the correlations were statistically significantly different from zero (language comprehension: Pearson $r[20] = .11$, $p = .61$, 95% CI $[-0.32, +0.51]$; word production: Pearson $r = -.09$, $p = .68$, 95% CI $[-0.49, 0.34]$; sentence production: Pearson $r = .014$, $p = .95$, 95% CI $[-0.42, 0.44]$).

Discussion

Previous studies on turn-taking in children with DLD suggested that the reduced number of conversational turns in children with DLD (Blom et al., 2023; Hadley & Rice, 1991; Mohan et al., 2021) is not likely to be explained by difficulties with general pragmatic aspects of turn-taking, such as interrupting the conversational partner to an unusual degree, differences in topic control, or

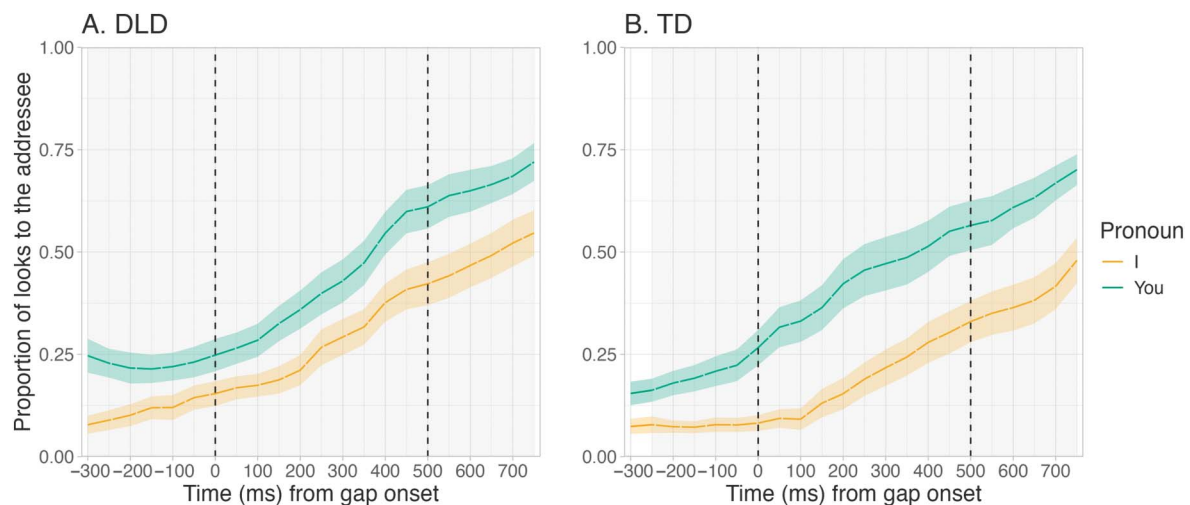
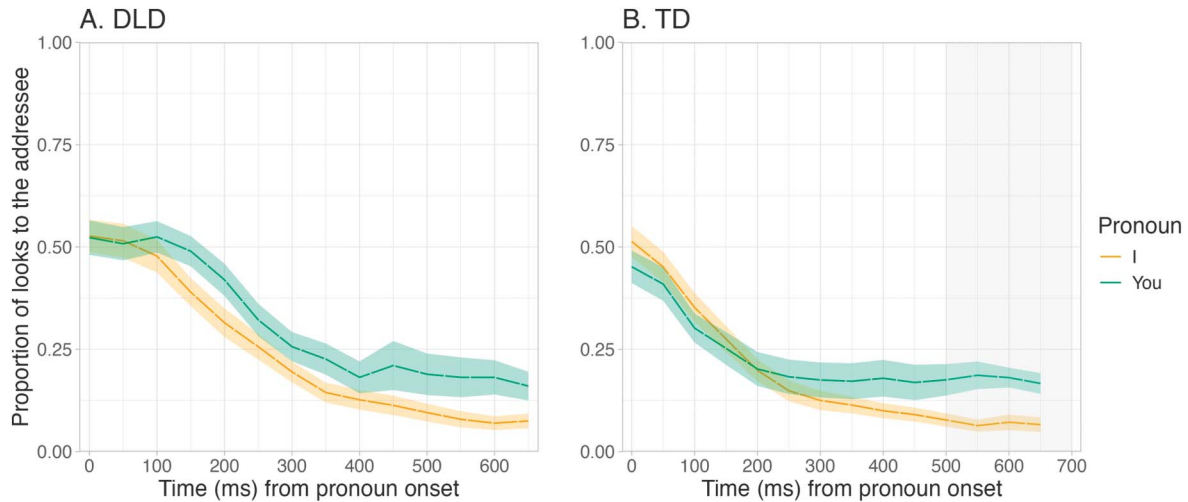
Figure 7. Children's proportion of looks to the addressee for questions with the second-person pronoun (green) and questions with the first-person pronoun (orange) from 300 ms before the gap to 300 ms after the gap. The area between the two solid lines (from 0 to 500 ms) indicates the 500-ms gap (interutterance silence). The light gray shading indicates time bins that were found to be statistically significant in the permutation analysis. (A) Gaze behavior of the children with developmental language disorder (DLD). (B) Gaze behavior of the typically developing (TD) children.

Figure 8. Children's proportion of looks to the addressee for questions with the second-person pronoun (green) and questions with the first-person pronoun (orange) in the 700 ms following pronoun onset. The light gray shading indicates time bins that were found to be statistically significant in the permutation analysis. (A) Gaze behavior of the children with developmental language disorder (DLD). (B) Gaze behavior of the typically developing (TD) children.



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not providing any response (Bishop & Adams, 1989; Justice & Kaderavek, 2003; Loveland et al., 1988; Moseley, 1990; Rice et al., 1991; Stojanovik, 2006; van Balkom et al., 2010, but see Craig & Evans, 1989; Mohan et al., 2021). In the present study, we therefore explored a novel, alternative explanation, namely, that relative lower language proficiency in children with DLD hinders not only children's understanding of utterances but also their real-time integration of fleeting linguistic cues to upcoming questionhood—cues that are essential for maintaining a smooth conversational flow. Smooth turn-taking requires the ability to predict if and when a response is needed. Certain speech acts, such as greetings and questions, increase the likelihood of a needed response. These speech acts are often linguistically marked (e.g., Dutch yes/no questions have subject-verb inversion and rising intonation; questions are more likely to have a second-person pronoun than a first-person pronoun). Sensitivity to such markers may facilitate the prediction of upcoming conversational structure. Indeed, from 2 years of age, TD children showed a question advantage and a pronoun advantage in their anticipation of upcoming turn transitions. That is, from this age onwards, children anticipated more speaker changes for questions than nonquestions (Casillas & Frank, 2017) and for questions with the second-person pronoun than for questions with the first-person pronoun (Lammertink et al., 2025).

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Children with DLD have difficulties in processing linguistic information, so in the present study, we investigated whether their difficulties with processing linguistic information (negatively) impact their prediction of upcoming conversational structure. In doing so, we investigated whether 3-year-old Dutch children with suspected DLD

show similar question and pronoun advantages in their real-time prediction of upcoming turn transitions, compared to their TD age-matched peers. Second, for the children with suspected DLD, we also explored whether their scores on language tests correlated with the size of their question or pronoun advantage.

For the question advantage, we found no evidence that the size of the question advantage differs between children with and without suspected DLD. That is, both groups of children anticipated more speaker switches for questions than nonquestions. However, in their timing of looks to the addressee for questions versus nonquestions, the two groups of children did seem to differ: Our cluster-based permutation analysis suggests that TD children started to switch their gaze from the speaking character to her addressee earlier on; TD children looked more at the addressee for questions than nonquestions while the speaking character is still talking, whereas children with suspected DLD started to look more at the addressee for questions than nonquestions just after the speaking character finished their turn (i.e., 150 ms into the gap). Given that the subject-verb inversion cue—one of the cues to questionhood present in our stimuli—is available right from the start of the utterance, we also looked at children's proportion of looks in a time window following 800 ms after the start of our target utterances. Here, we found that TD children (temporarily) started to look more at the addressee for questions versus nonquestions 500 ms after utterance onset, suggesting that they indeed use subject-verb inversion as an early cue to questionhood and conversation structure in unfolding conversations. We had no evidence for such early cue use in children with DLD.

The outcome that children with DLD do predict upcoming (linguistic) material during real-time sentence processing, but more slowly than TD children, may be more consequential in natural first-party conversations than in our experimental study, which only measures anticipation during third-party observation. In natural conversations, the silences between two speaker turns are typically shorter than 500 ms (at least in adult–adult conversation; Stivers et al., 2009). Actual conversational turn-taking is also more variable in the available linguistic and nonlinguistic cues (Holler & Kendrick, 2015) and may require children to formulate and articulate a response in overlap with their comprehension of unfolding speech (Casillas & Frank, 2017; Levinson, 2016). The psycholinguistic demands of turn-taking in real, first-party conversation help to explain why children tend to respond more slowly than adults (e.g., Garvey & Berninger, 1981; Stivers et al., 2018), particularly when questions and answers present challenges (Casillas et al., 2016; Lindsay et al., 2019). Our findings also fit with studies on predictive processing in children with and without (suspected) DLD at the sentence level. For example, in a recent study by van Alphen et al. (2021), 3-year-old children with and without suspected DLD performed a looking-while-listening task. During the experimental trials, children saw two pictures on a screen (e.g., a book and a sock) while they listened to utterances. The verb in the unfolding utterance was either semantically related to the last word of the utterance (e.g., he reads a book) or semantically neutral (he sees a book). Comparing the anticipatory looks to the target pictures, van Alphen et al. (2021) observed that the 3-year-old children with suspected DLD anticipated target pictures more slowly upon hearing semantically related verbs (e.g., reading) as compared to their TD age-matched peers.

Note, however, that the use of linguistic cues to anticipate upcoming information in children with DLD may depend on the type of linguistic cue. Deevy and Leonard (2018), for example, found no evidence that 5-year-old children with DLD used tense agreement cues to anticipate a target picture in fronted auxiliary questions (e.g., “Are/Is the nice little dog/s running?”) while 3-year-old TD children already did. The younger children with DLD in the present study seemed to have used the linguistic cues (subject–verb inversion and rising intonation) to recognize questions and to anticipate upcoming speaker changes, albeit at a slower pace than their TD age-matched peers. One possible, and speculative, explanation for their slower pace is that the children with DLD mainly relied on the prosodic cue (rising intonation) to questionhood. This prosodic cue became available at the end of the utterance—much later than the syntactic cue (subject–verb inversion). This explanation warrants further investigation, given that our current design cannot disentangle children’s use of syntactic and prosodic cues to questionhood.

The outcomes for the pronoun advantage are a bit harder to interpret. If we compare the proportion of anticipatory looks for questions with the second-person pronoun versus questions with the first-person pronoun, then we observe the expected pronoun advantage in both groups of children. Similarly to TD children, by the age of 3 years, children with suspected DLD use the pronoun cue to anticipate upcoming conversational structure (Lammertink et al., 2025). That is, at first sight, children with DLD appear as efficient as their TD peers in their use of this cue: Both groups of children, after hearing a second-person pronoun, start to look more at the addressee while the speaking character is still talking (i.e., 250–300 ms before the end of their turn). However, when we examined an earlier window, in our second permutation analyses, we found that TD children indeed already started to look more at the addressee approximately 500 ms after pronoun onset, whereas we found no evidence for statistically significant time bins following the 700 ms after pronoun onset in children with DLD. From this pattern of findings, we speculate that children with DLD understand the heuristic, question-associated implications of second-person pronoun use; however, as with cues to questionhood (e.g., subject–verb inversion), they require more time to process those implications and apply them in real-time prediction. Similar conclusions have been drawn regarding children with DLD’s understanding of binding principles (i.e., linking pronouns to their structurally licensed antecedent) during real-time sentence processing (Abbott et al., 2024). That said, we cannot rule out the possibility that the power of our study was too low to detect a difference.

In a final step of the analyses, we explored correlations between children’s language proficiencies and their proportion of anticipatory gaze switches in relation to the size of children’s question and pronoun advantages. All the correlations fell within the small-to-moderate range ($r = .05-.40$), and none was statistically significantly different from zero. This finding was somewhat unexpected given that correlations between predictive processing (at the sentence level) and language proficiency have been observed in previous studies with TD children (Kamide et al., 2003; Mani & Huettig, 2012). Although the absence of correlations is a null effect and therefore difficult to interpret, we would like to present one possible explanation for the observed pattern. In our study, the largest correlation was observed between the size of children’s question advantage and their language comprehension scores. Our group of children with suspected DLD demonstrated relatively strong language comprehension skills; at the group level, their mean percentile score for language comprehension (TBQ = 92) actually fell within the TD range (85–115). Additionally, our task involved passively observing conversations, which emphasizes language comprehension and processing rather than active production.

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The relatively high correlation between language comprehension and the size of the question advantage in our study may thus reflect that our task is more an indication of children's language comprehension skills rather than their language production skills. In real-life conversations, where children are required to formulate and articulate responses, language production skills likely play a greater role (Casillas et al., 2016; Levinson, 2016; Lindsay et al., 2019). In future work, it would therefore be interesting to examine the role of language production and the anticipation of conversational structure in children with DLD more specifically.

Alternatively, children with both comprehension and production difficulties could be compared to those with production difficulties only, given that the children with both comprehension and production difficulties have been found to have more difficulties with conversation and turn-taking, compared to children with production difficulties only (Craig & Evans, 1993). Some have suggested that difficulties with utterance formulation explain the lack of responsiveness sometimes seen in conversations with children with DLD (Hadley & Rice, 1991) and that children with language difficulties compensate for their difficulties with utterance formulation using nonverbal cues such as gestures (Mohan et al., 2021). However, others (e.g., Bishop et al., 2000) argue that difficulties with utterance formulation cannot explain the lack of responsiveness in children with language difficulties, given that similar nonresponses are also seen in acknowledgment-soliciting utterances, which require only that the child nods for agreement (Bishop et al., 2000).

Limitations and Future Studies

In this study, we conducted a group-level comparison between children with suspected DLD and their age-matched peers to further understand the processing of linguistic cues in the anticipation of conversational structure in children with suspected DLD. Three aspects in this group-level comparison warrant further discussion. First, reviewers raised the possibility that including three multilingual children in our sample of children with suspected DLD—but not any TD multilingual children—could have influenced the results. Given cross-linguistic differences in the use of our target linguistic cues to mark yes/no questions (i.e., subject–verb inversion, rising intonation, and subject pronoun), one could hypothesize that the three multilingual children may have been at a disadvantage when processing these cues because they may have been exposed to the linguistic cues less frequently than their monolingual peers, due to their mixed language input. However, given the high frequency of subject–verb inversion and the association with the second-person pronoun in Dutch yes/no questions (see the supplementary analyses in Lammertink et al.,

2025), and given that Dutch was one of the primary languages for all three multilingual children, we would expect them to be quite familiar with these cues. Alternatively, the differences in processing cues between multilingual and monolingual children may be due to the multilingual children's relatively low Dutch language proficiency, as compared to their monolingual peers. Due to their mixed language input, the Dutch proficiency of multilingual children with suspected DLD may be lower than that of monolingual children with suspected DLD. Indeed, when we compared the Schlichting test scores between multilingual and monolingual children with suspected DLD, we found that the former had statistically significantly lower language comprehension scores. We found no evidence of differences in language production scores between the two groups. From this pattern, we cannot claim that reduced Dutch input causes the multilingual children's DLD or their generally lower proficiency (see also Kohnert, 2010; Peña et al., 2023). We also realize that these standardized test scores for the multilingual children should be interpreted with caution, given that we have no information about how long the multilingual children have been exposed to Dutch or which language is their most dominant language (i.e., because Dutch was one of their primary languages but not their most dominant; Paradis, 2010). That being said, the lower language comprehension scores of these children may have impacted our results, given our observed trend for a moderate correlation between children's language comprehension scores and the size of their question advantage, and that anticipating conversational structure requires the integration of multiple linguistic cues in real time. For these reasons, we ran additional post hoc analyses to explore whether our findings change when we exclude the three multilingual children with suspected DLD. To do this, we added language status (multilingual vs. monolingual) as a fixed effect to our generalized linear mixed-effects models, which assessed the size of the question and pronoun advantages. Post hoc analyses provided no evidence that including multilingual children affected the size of the question advantage, nor the group difference in the question advantage between children with and without suspected DLD. Furthermore, while including the three multilingual children may have reduced the overall size of the pronoun advantage, there is no evidence to suggest that this affected the size of the difference in the pronoun advantage between children with and without suspected DLD.

Second, for practical reasons, we had to use two different eye trackers for our participant groups. We do not know whether these tracker differences drove any of our findings, but if so, we expect the impact to be minimal, given the characteristics of our design (i.e., large areas of interest) and the age of our participants (differences between eye trackers have been reported to affect data

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quality more in infants than in toddlers; de Kloe et al., 2021). We are also not the first to use different eye trackers in one data set: Several of the large-scale replication projects, for example, also include data from different laboratories using different eye trackers (e.g., Byers-Heinlein et al., 2021; Spit et al., 2023). Nevertheless, it may be worthwhile for child language researchers to further explore whether and how the use of different eye trackers may impact study outcomes. This is also why planned projects such as the ManyBabies-ManyTrackers project are important (<https://manybabies.org/MB-ManyTrackers/>). Third, the number of children tested in both groups is relatively small, which may have resulted in power too low to detect differences between the groups of children, particularly given that the effect sizes in both the question and pronoun advantages are relatively small.

Conclusions

The outcomes of this study show that, at 3 years of age, children with and without suspected DLD have sufficient linguistic knowledge to distinguish questions from nonquestions and that they use linguistic information (subject–verb inversion, rising intonation, and subject pronoun) in their real-time prediction of conversational structure. However, the children with suspected DLD in our study were less efficient in their predictions of upcoming speaker change. The results suggest that the difficulties with turn-taking sometimes observed in children with DLD may be related to slower processing of linguistic information, particularly children's use of this information to predict an upcoming speaker change in real time.

Data Availability Statement

The data sets generated and analyzed during the current study are available in the Radboud Data Sharing Collection (Lammertink, Rowland, & Casillas, 2025).

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